

16/07/24:-

CER 3G1 → computer organization and Architecture

Date: / / Page:

Computer

↳ Hardware/
Software

↳ Need program/
software to
perform calc.

↳ It is a programmable
machine.

Calculator

↳ Hardware/
firmware

↳ don't need
any software

↳ It is a simple
Hardware Machine
with some limitations

↳ All computer → Same design → different components

↳ i3 differ from i7 → Because of Hardware present →
increases the ability & analyzing Power

↳ Program/code error → syntax error

↳ Computer Memories/ → ① RAM ② Pendrive
storage
↳ 8GB & 16GB
↳ 32GB & 64GB

30/July/24:-

Computer Organization & Architecture:- (Book name)
↳ By William Stallings (10th ed.)
(suferred Book)

◎ Boolean → Powerful mathematics, which gives the evolution of digital system. (in Binary)

↳ machine language (in Binary format)
↳ Assembly language
↳ ~~XXXXXX~~ C
↳ 4GLS

} processor dependent

} processor independent
↳ as we go downwards,
 ~~coding~~ coding implemented in
 english language

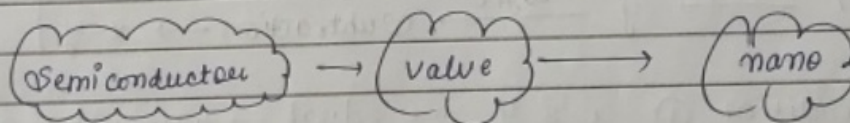
(HDD → Hardware device) ← PC

SSD (Semiconductor device)
mechanical devices

Both are memory organizer

But SSD is faster 'cause of Semiconductor

↳ Architecture is the logical part, how does it works (organises)
↳ organizer is a organised part



↳ use of biochips → No connectivity, but it transform signals. (it will come in future)

↳ certain chemical present → Brain perform → Logic gates

↳ Logic gates used → as a building block while studying mathematical de-morgan's law, Arithmetic op.
 ↳ +, -, *, ÷

→ using X OR GATE $\xrightarrow{\text{perform}}$ 2 operation $\rightarrow +, -$

→ By using repetitive Addition, we can $\rightarrow *$ perform

→ " " " subtraction " $\rightarrow \div$

{ Basic architecture $\rightarrow$ particularly organized $\rightarrow$ we can designed $\rightarrow$ a perfect device }

ALU $\rightarrow$ Arithmetic Logical unit

if } $\rightarrow$ single statement $\rightarrow$ compare & Jump
else }

cmp (instruction)

$\begin{matrix} < > < > \\ \downarrow & \downarrow \\ \text{Register ①} & \text{Register ②} \end{matrix}$ } $\rightarrow$ compare Register ① & Register ②

Accumulate

$\left[x \xrightarrow{\text{compare}} y \xrightarrow{\text{Logic}} \text{Subtraction} \rightarrow x - y \right]$

→ condition $\left. \begin{array}{l} \text{① } x \& y \text{ are equal} \rightarrow \text{result} = 0 \\ \text{② } x > y \rightarrow \text{result } (+) \\ \text{③ } x < y \rightarrow \text{result } (-) \end{array} \right\}$

part of Architecture

→ CPU used in computers $\left. \begin{array}{l} \rightarrow \text{intel processor} \\ \rightarrow \text{mac processor} \end{array} \right\} \rightarrow$ sudden partition must be differ.

- Architectural difference
- organisational difference

→ in intel & mac processor

mac → RISC (Reduced Instruction set computer)
 intel → CISC

RISC → Less NO. of instructions

CISC → more NO. of instructions

→ Android using ARM Processors
 Mobile users

→ peripherals, i/o devices

→ How they were connected
 → " " " used

In systems:-

[power supply doesn't affect the performance → power supply (Mechanical device)]

KB → mechanical device

then, Hard-disk, SSD

→ Instruction cycle

- Types of Instruction
- How Instruction works
- What is Instruction

Study

→ Optimization is always required

→ in contrast of Speed and technology

* Always used variable → acc-to your compiler
 → memory.

memory → 0mb — 64kb